Lecture 10

Representation Learning and Information Theory

Speaker: Kaiming He

Overview

- Basics of Information Theory

- Modeling Information with Neural Networks

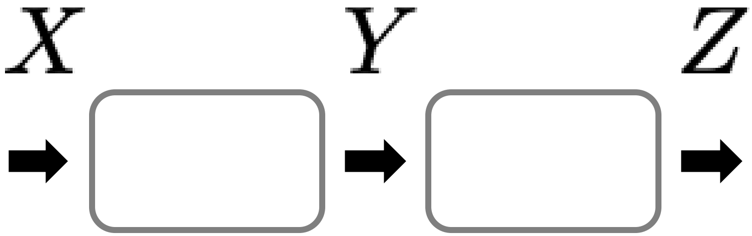
What are “good” representations?

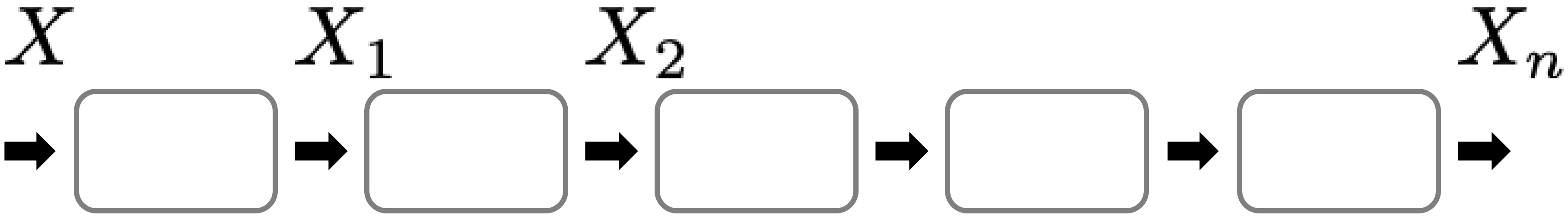
We may want representations that:

- retain *information* about the input
- discard *information* irrelevant to the tasks
- compress *information* effectively under limited capacity

But:
- What is “information”?
- How can we measure information?
- with neural networks?

In the context of neural nets:





We may ask questions such as:

- What is the mutual information between X and Z ?
- How much information is lost in X_n vs. X_2 ?
- Is X_n more informative than Z ?
- …

Basics of Information Theory

See also:
6.7480 Information Theory

Information Measures

- **Entropy**: measure of uncertainty

$$H(X) = -\sum_{x \in \mathcal{X}} p(x) \log p(x) = -\mathbb{E}_{x \sim p} \log p(x)$$

X : a random discrete variable

$\mathcal{X}$: the support set e.g. $\mathcal{X} = \{0, 1, 2, 3\}$

x : a realization e.g. $x = 2$

*We discuss discrete variables in this lecture for simplicity

Information Measures

- **Entropy**: measure of uncertainty

$$H(X) = -\sum_{x \in \mathcal{X}} p(x) \log p(x) = -\mathbb{E}_{x \sim p} \log p(x)$$

most uncertain

Example: Binary entropy $\mathcal{X} = \{0, 1\}$

$$\text{Prob}(X{=}1) \triangleq p$$

$$H(X) = -p \log p - (1-p) \log(1-p)$$

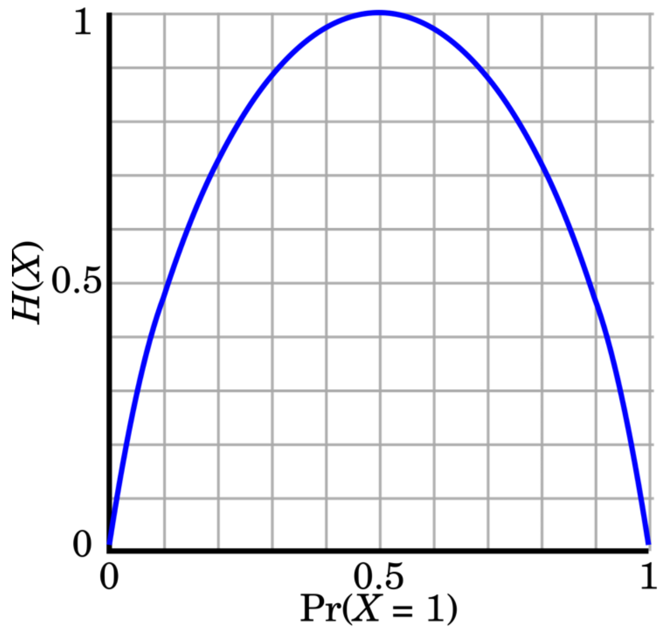


figure: https://en.wikipedia.org/wiki/Binary_entropy_function

Information Measures

- **Joint entropy**: uncertainty of joint distribution

$$H(X, Y) = -\mathbb{E}_{p(x,y)} \log p(x, y)$$

- **Conditional entropy**: uncertainty of X given Y

$$\begin{aligned} H(X \mid Y) &= \mathbb{E}_{p(y)} H(X \mid Y = y) \\ &= -\mathbb{E}_{p(x,y)} \log p(x \mid y) \end{aligned}$$

Information Measures

- **Mutual Information**: information provided by X about Y

$$I(X;Y) = \mathbb{E}_{p(x,y)} \log \frac{p(x,y)}{p(x)p(y)} = D_{\mathrm{KL}}(p(x,y) \,||\, p(x)p(y))$$

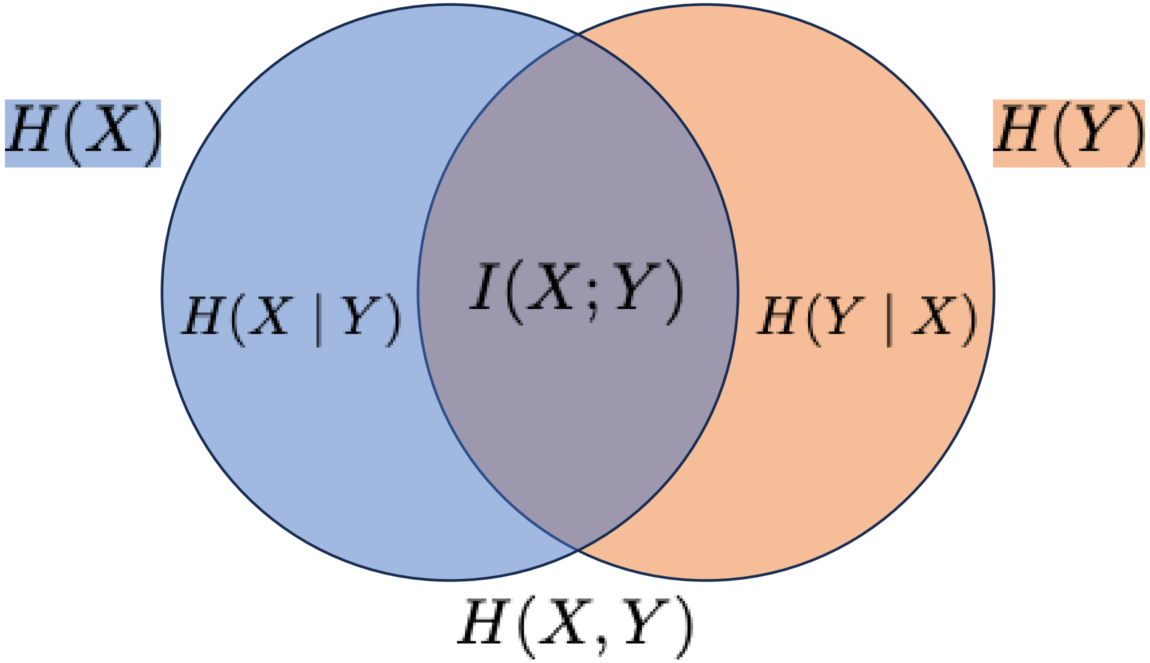
Properties:

$I(X;Y) = I(Y;X)$	symmetry
$I(X;Y) = H(X) - H(X \mid Y)$	reduction in uncertainty of X given Y
$I(X;Y) = H(Y) - H(Y \mid X)$	reduction in uncertainty of Y given X

Information Measures

$$I(X;Y) = H(X) - H(X \mid Y)$$

$$I(X;Y) = H(Y) - H(Y \mid X)$$

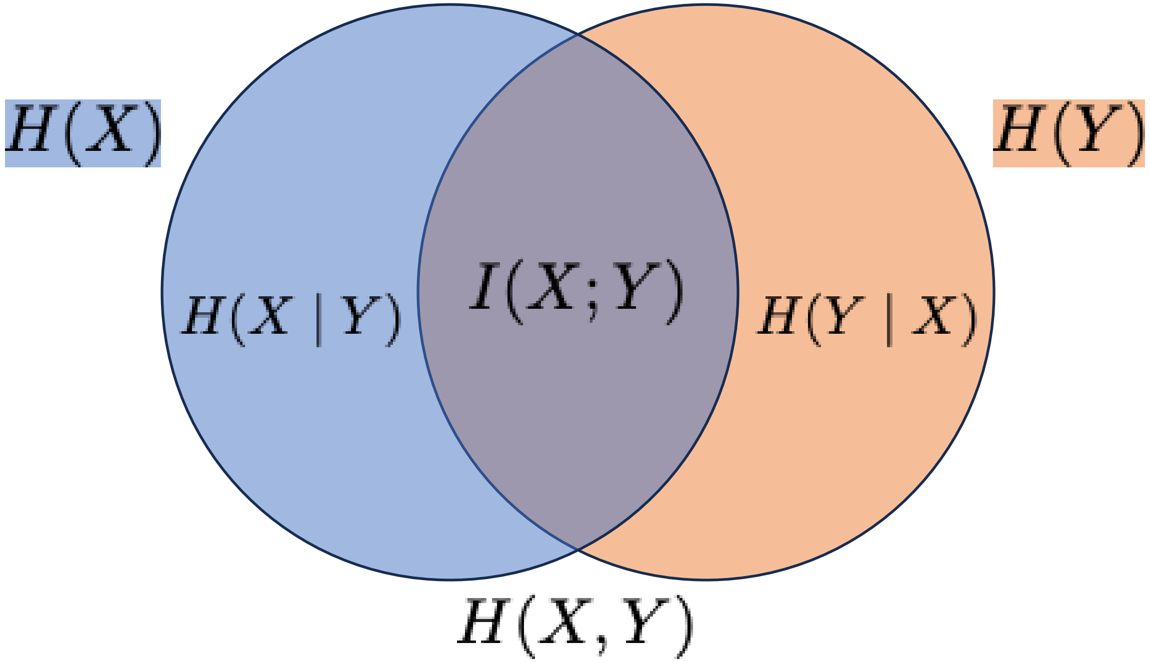


$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

figure adapted from: https://en.wikipedia.org/wiki/Mutual_information

Information Measures

$I(X;Y) \le H(X)$ $\qquad$ $I(X;Y) \le H(Y)$



non-negative:

$$I(X;Y) \ge 0$$

$$I(X;Y) = 0 \iff X \perp Y$$

figure adapted from: https://en.wikipedia.org/wiki/Mutual_information

Information about X cannot increase through processing

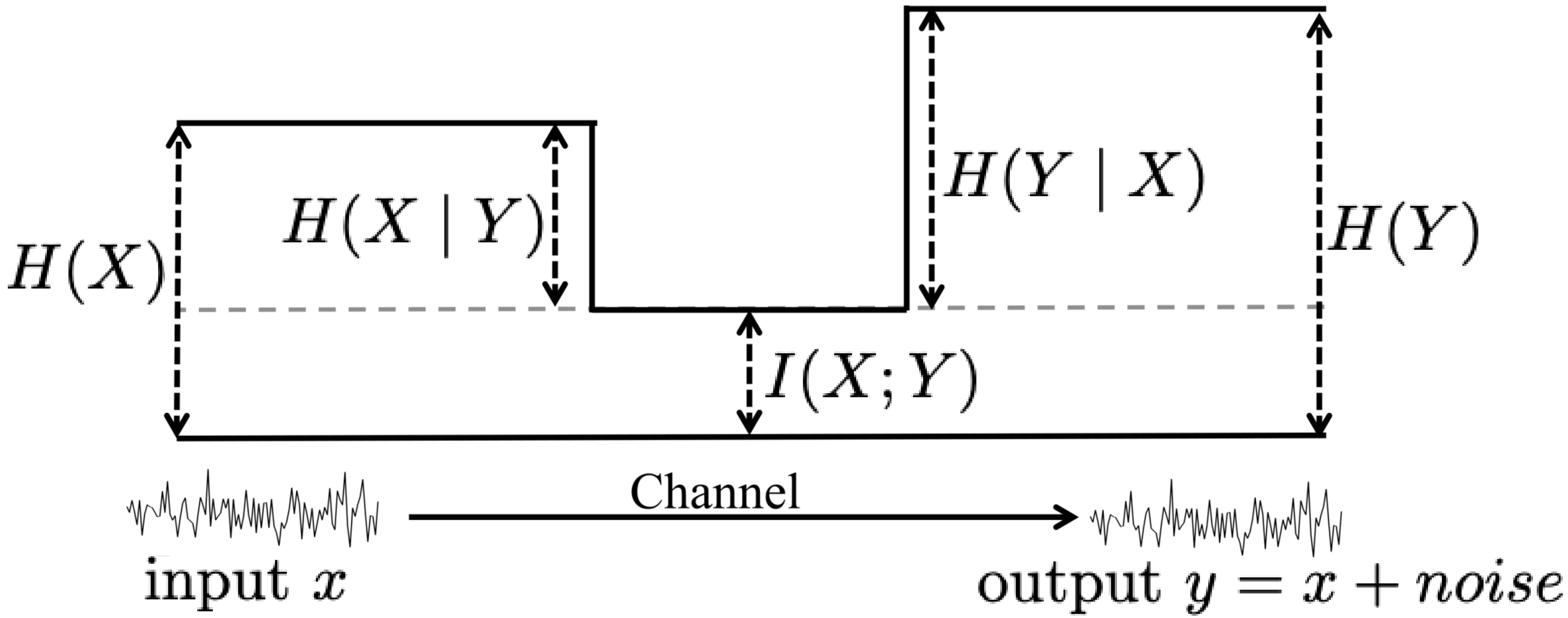
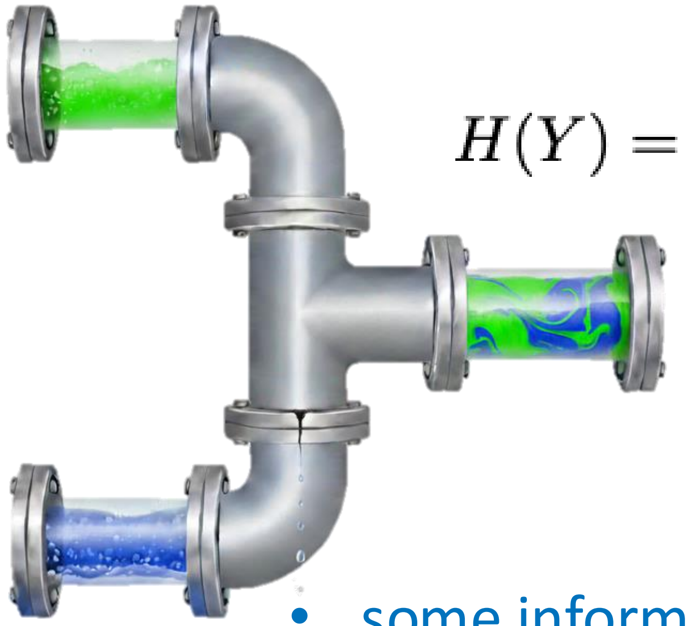


figure adapted from: https://en.wikipedia.org/wiki/Mutual_information

Information about X cannot increase through processing

noise



$$H(Y) = I(X;Y) + H(Y \mid X)$$

Y

$H(X)$

X

- information about X ("blue water") cannot increase:

$$H(X) \geq I(X;Y)$$

- some information can be lost ("water leak")

$$H(X \mid Y) \geq 0$$

illustration generated by Nano Banana (Gemini)

Information about X cannot increase through processing

$$X \rightarrow \boxed{\text{layer(s)}} \rightarrow Y$$

$$H(X) \qquad\qquad I(X;Y)$$

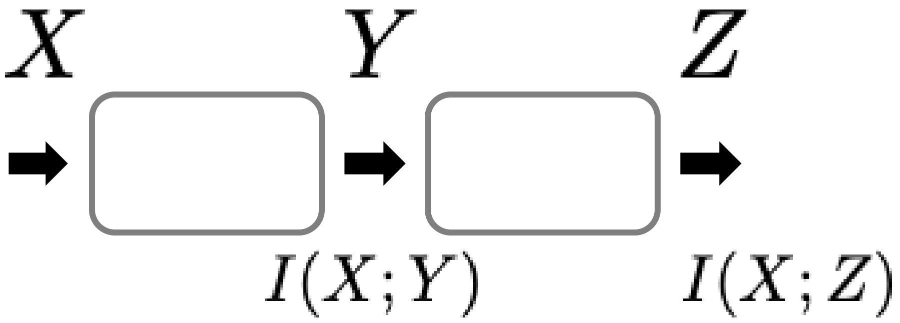
$$H(X) \geq I(X;Y)$$

Any network layer(s) can only
reduce or keep information about its input

Data Processing Inequality

Consider a Markov Chain: $X \rightarrow Y \rightarrow Z$

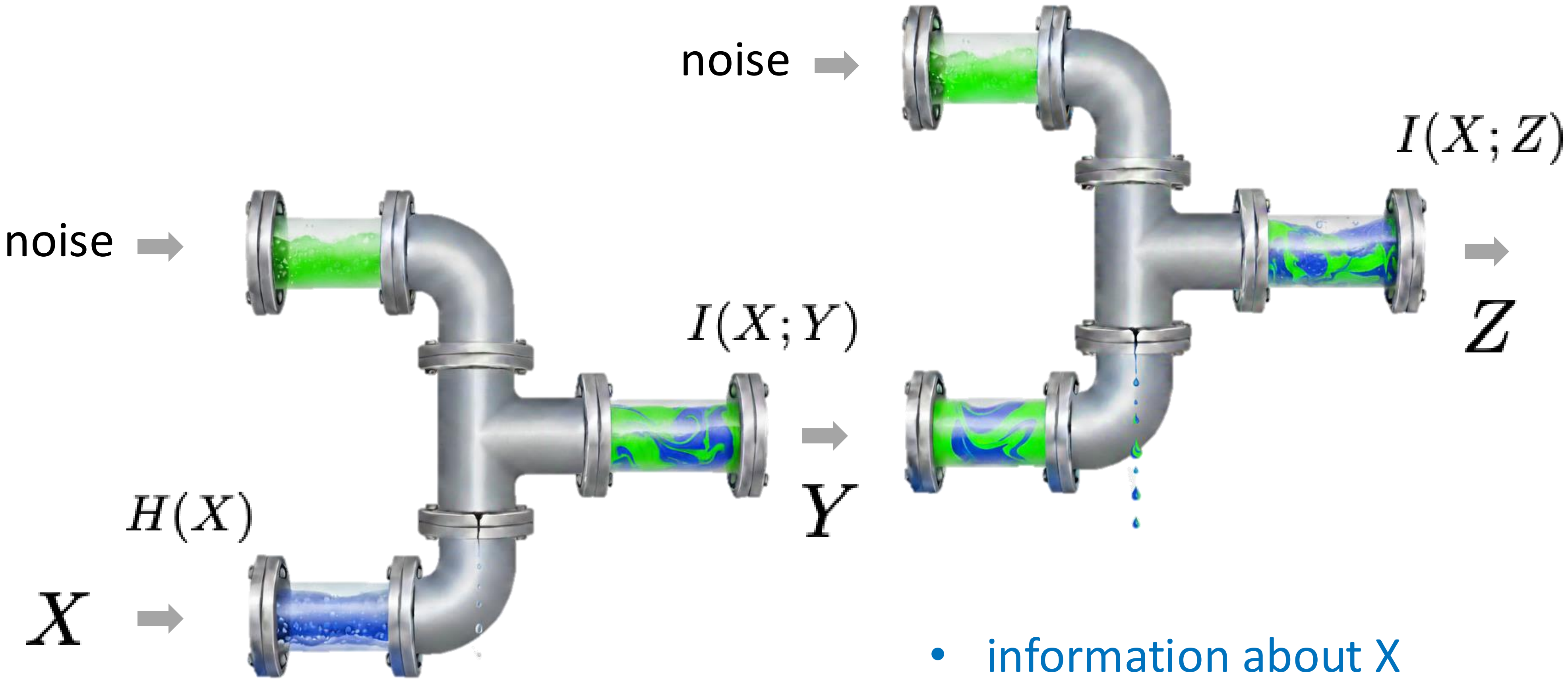
we have: $I(X;Y) \geq I(X;Z)$



$$I(X;Y) \geq I(X;Z)$$

Information about the input degrades or stays the same
with further processing.

Data Processing Inequality

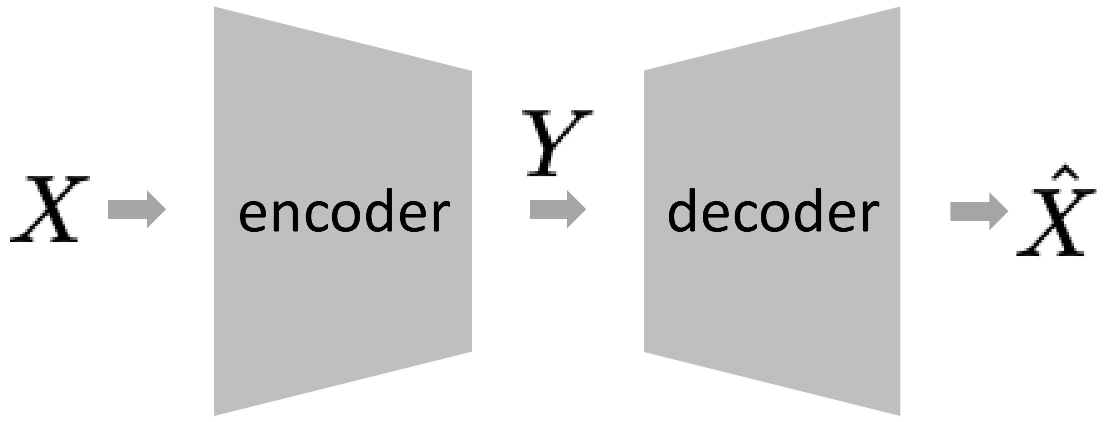


- information about X ("blue water") cannot increase:

$$H(X) \geq I(X;Y) \geq I(X;Z)$$

illustration generated by Nano Banana (Gemini)

Rate-Distortion Theory

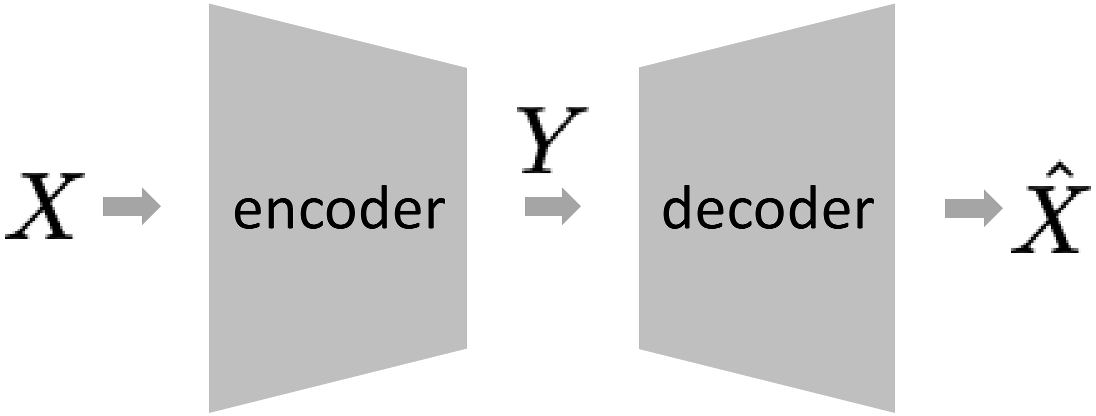


distortion measure: $d(x, \hat{x}) \geq 0$ e.g., Mean Squared Error (MSE)

rate: R average bits required to represent per realization

rate-distortion function: $R(D)$ minimum rate needed such that the expected distortion does not exceed D

Rate-Distortion Theory



$$R(D) = \inf_{P(\hat{X}|X):\mathbb{E}[d(X,\hat{X})]\leq D} I(X;\hat{X})$$

- lossless compression: $D = 0, R(D) = H(X)$
- As D increases, R decreases
- If D is infinite, R is 0

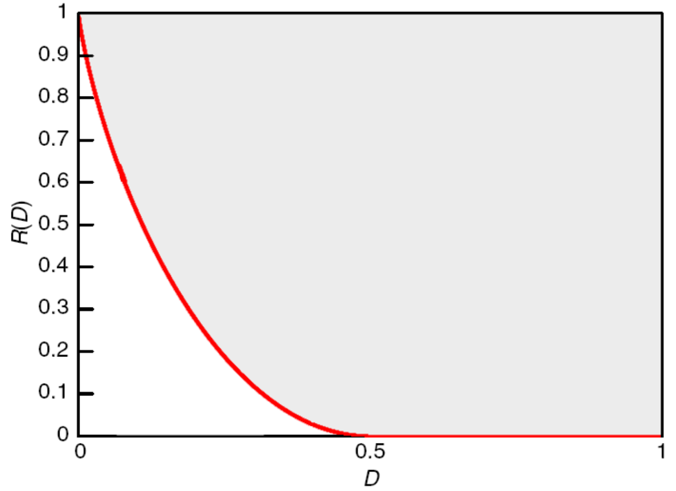
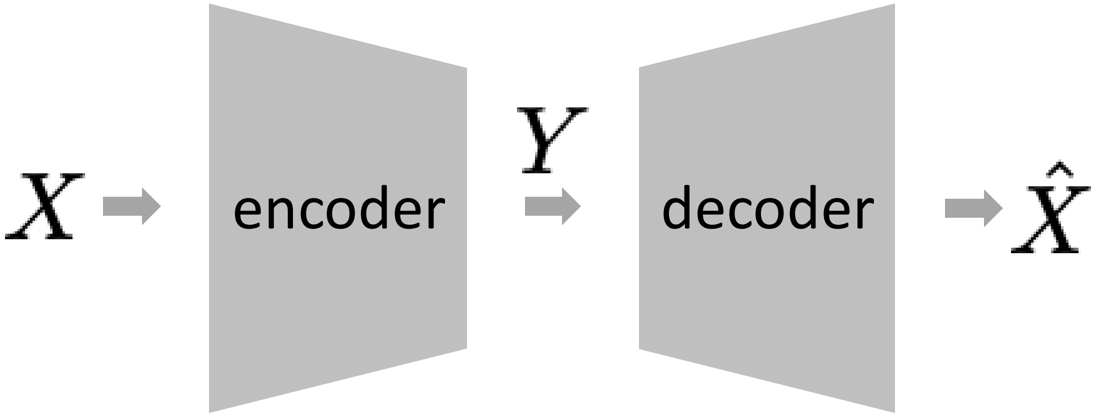


figure: https://en.wikipedia.org/wiki/Rate%E2%80%93distortion_theory

Rate-Distortion Theory

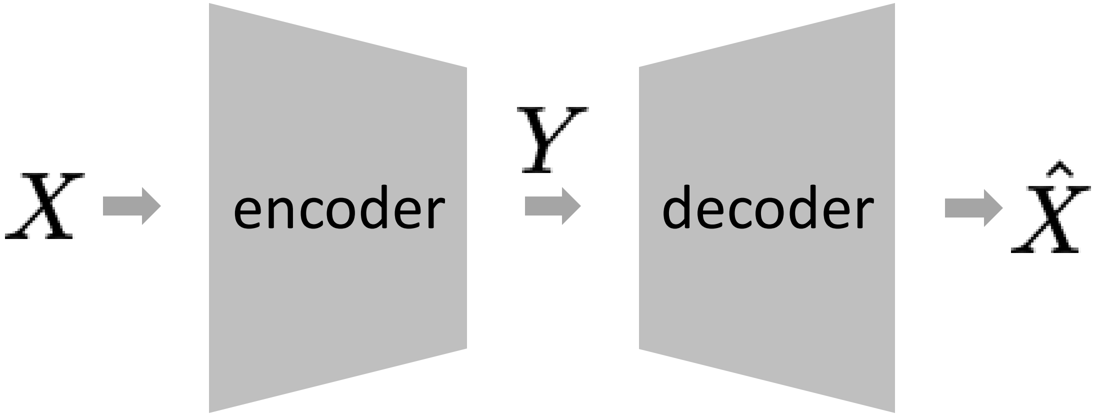


$$R(D) = \inf_{P(\hat{X}|X):\mathbb{E}[d(X,\hat{X})]\leq D} I(X;\hat{X})$$

- General Converse

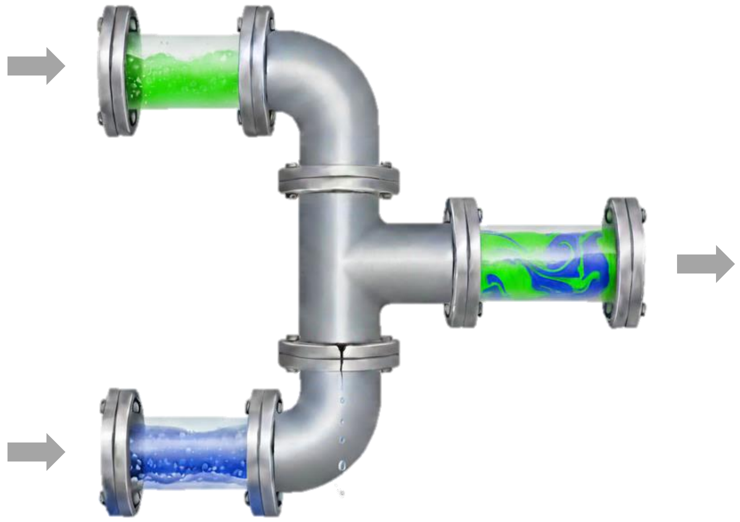
$$\text{“bits to encode } Y\text{”} \geq H(Y) \geq I(X;Y) \geq I(X;\hat{X}) \geq R(D)$$

If the “bottleneck” (in bits) is less than R(D), it would be impossible to reconstruct X within distortion D



It is the bottleneck that limits the maximum information going through a network.

Takeaways (informal)



- **Information** => "**water**"
- **Functions** => "**pipes**"

- "water" never increases in "pipes"
  - if multiple origins: "water" from each origin never increases

- "water" can leak out (which cannot recover)

- narrowest "pipe" (bottleneck) determines the maximum capacity of "water flow"

Modeling Information with Neural Networks

- The foundation of modeling information is the assumption of ***probability***.

- But where does ***probability*** come from?

- tl; dr: uncertainty.

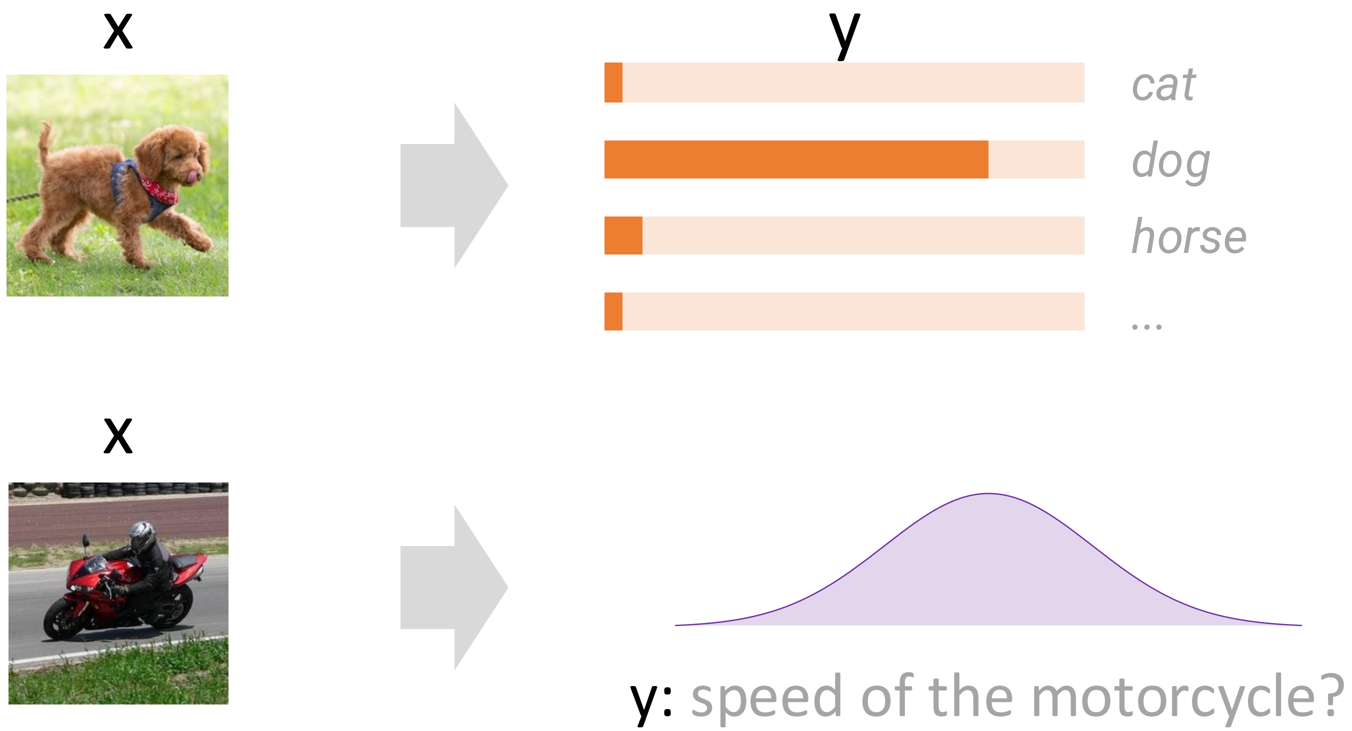
Where Does Probability Come From?

- X: an observation (e.g., images)
- Y: an attribute (e.g., class labels)

- Model $P(X,Y)$ as a joint distribution
  - Many plausible observations and attributes.
- Model $P(Y \mid X)$ as a conditional distribution
  - Given an observation, the attribute is uncertain
- Model $P(X \mid Y)$ as a conditional distribution
  - Given an attribute, the observation is uncertain

- Model $P(Y \mid X)$ as a conditional probability
  - Given an observation, the attribute is <u>uncertain</u>

discriminative models

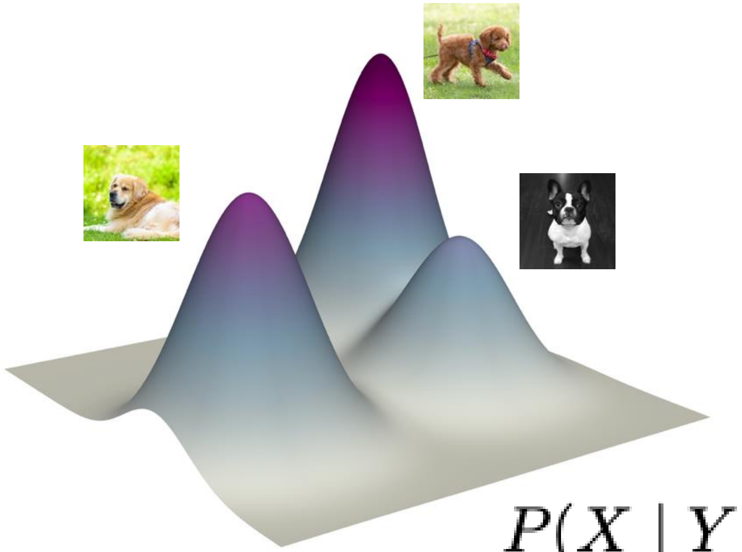


Uncertainty can arise from: noisy labels, imperfect models, or intrinsic ambiguity in the problem itself.

- Model $P(X \mid Y)$ as a conditional probability
  - Given an attribute, the observation is <u>uncertain</u>

generative models

$$Y = \text{“dog”}$$



$$P(X \mid Y = \text{“dog”})$$

- The “observation vs. attribute” distinction is **artificial**

- In general, X and Y can be **any variables**

Practical takeaway:
Think about **sources of uncertainty**

[Recap] What are "good" representations?

We may want representations that:

- retain *information* about the input
- discard *information* irrelevant to the tasks
- compress *information* effectively under limited capacity

An Information-theoretic Perspective on ...

- Supervised Learning
- Self-supervised Learning: Autoencoding
- Self-supervised Learning: Contrastive Learning

An Information-theoretic Perspective on ...

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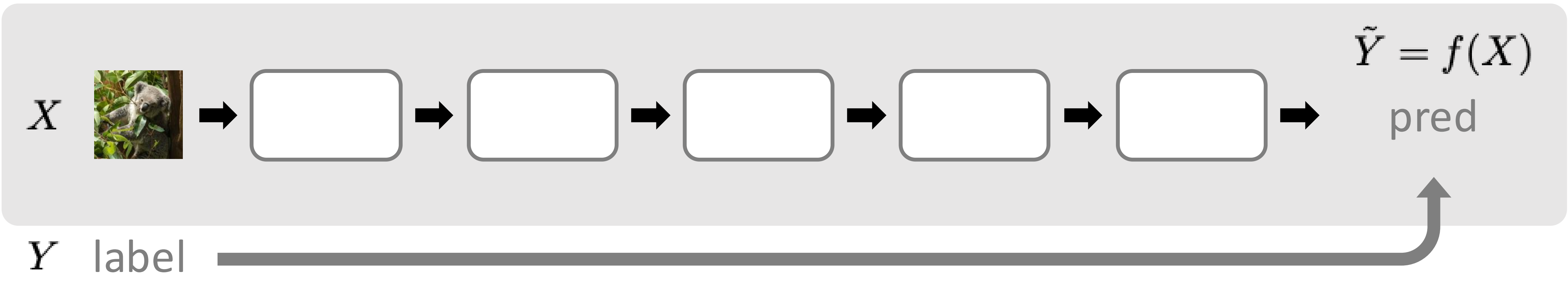
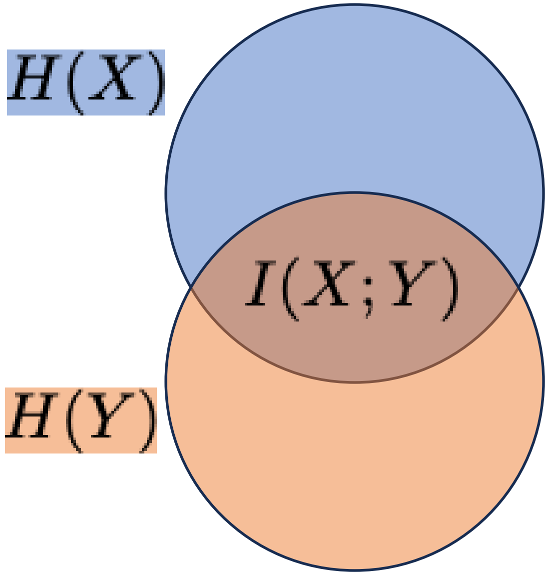
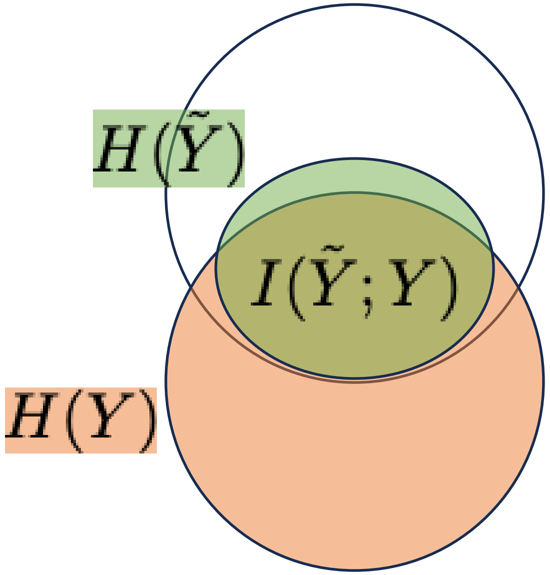
Supervised Learning



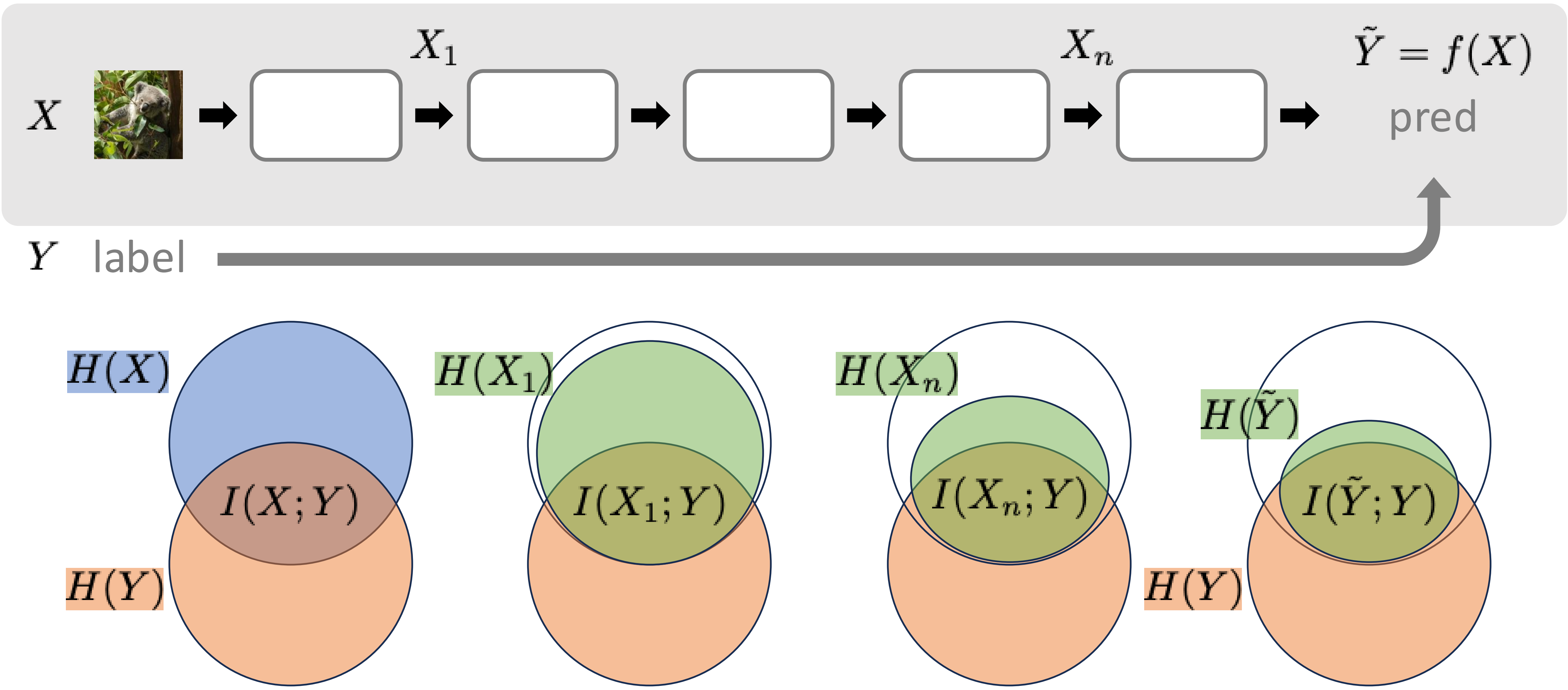


$$I(X;Y) \geq I(\tilde{Y};Y)$$

(Data Processing Inequality)

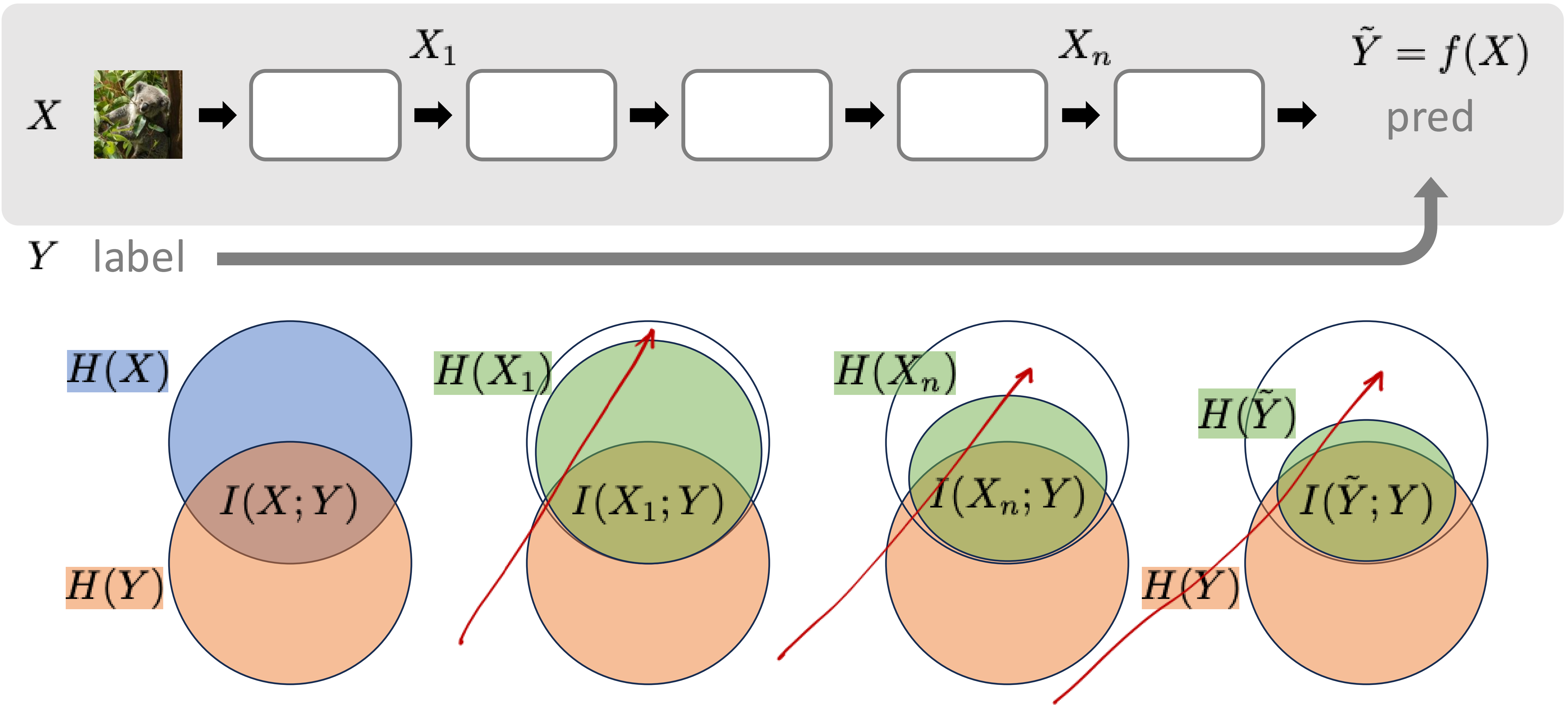


Supervised Learning



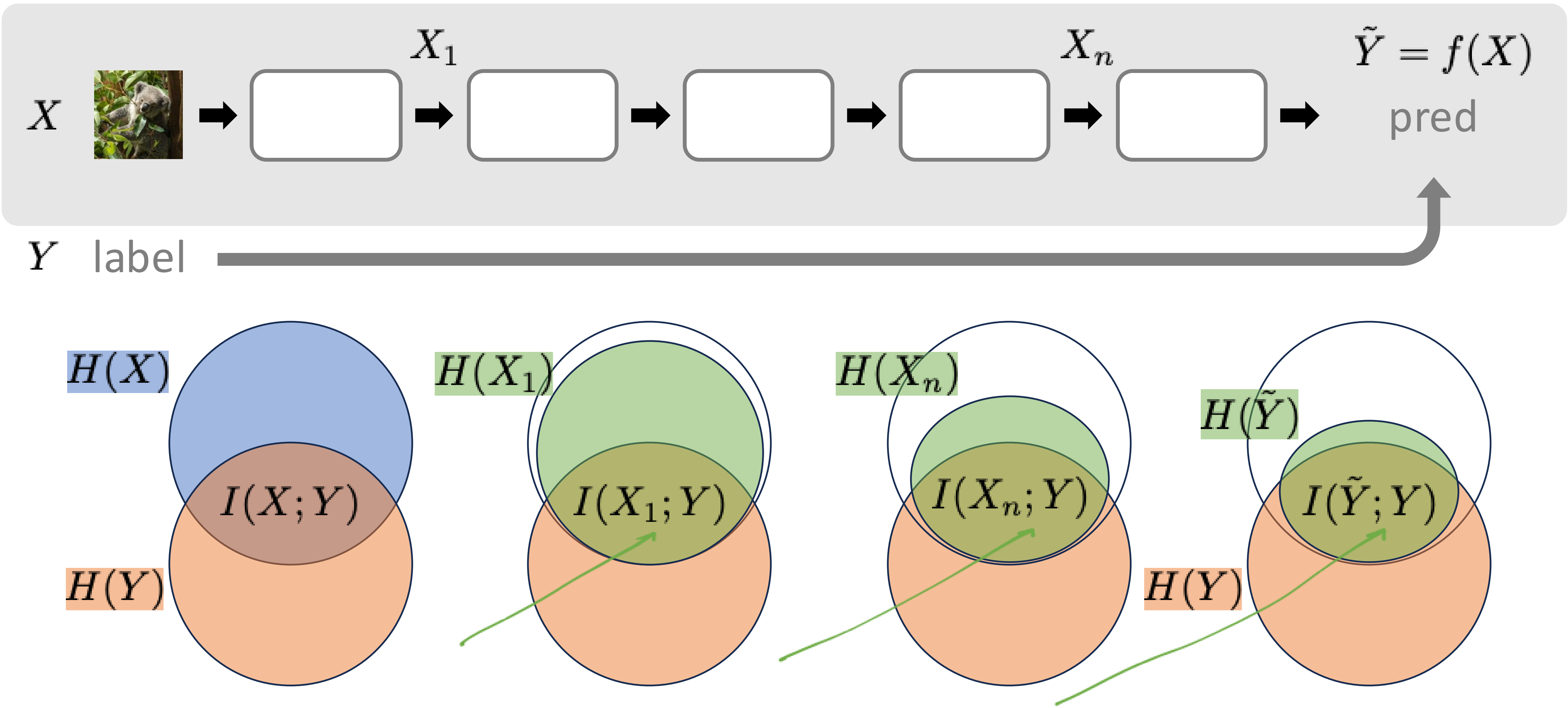
$$I(X;Y) \geq I(X_1;Y) \geq \ldots \geq I(X_n;Y) \geq I(\tilde{Y};Y)$$

Supervised Learning



- **Information discarded**: ideally task-irrelevant

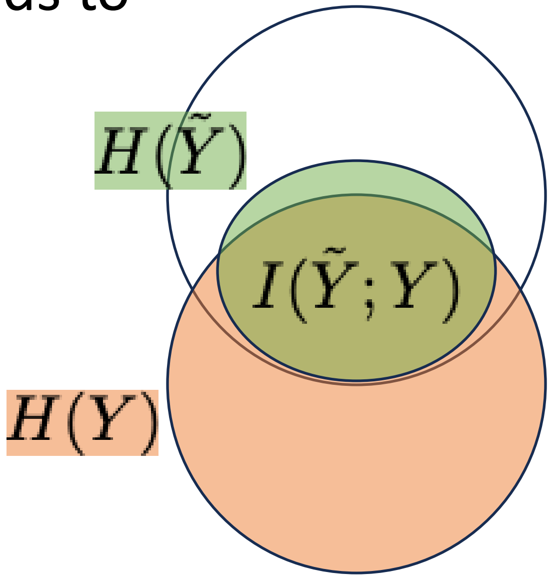
Supervised Learning



- **Information retained**: ideally only about the label Y

In Supervised Learning, ...

- $I(X;Y)$: the target information to retain
- $H(X \mid Y)$: ideally, the information to discard
- Cross-entropy loss encourages retaining $I(X;Y)$
  - if optimum is achievable, minimal cross-entropy loss leads to maximal mutual information $I(\tilde{Y};Y) = I(X;Y)$



What Makes Modeling Information Difficult?

- Analytical formulations
  - Only for limited families (Gaussian, uniform, categorical, ...)
  - low-dimensional
  - In supervised classification: categorical

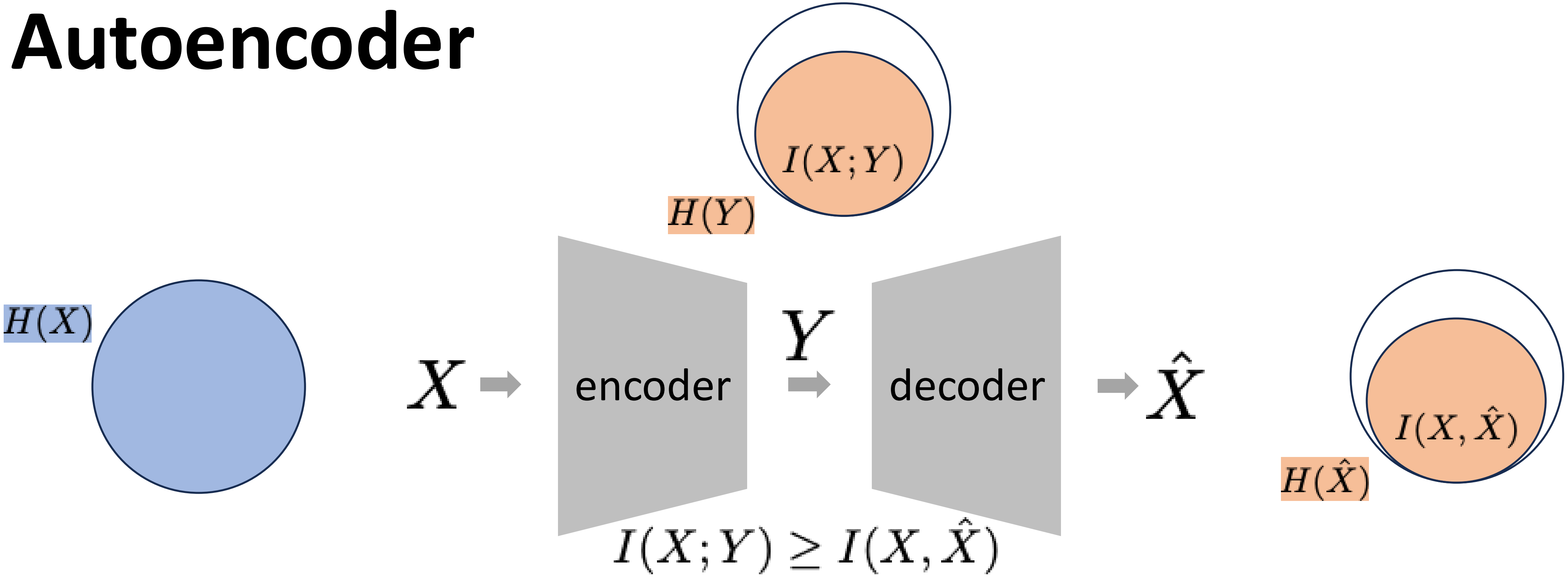
- Monte Carlo estimation
  - Distributions exist but unknown
  - We can only sample from the unknown distributions
  - In supervised classification: $x, y \sim p_{\text{data}}(x, y)$

- Information measures
  - often not directly computable (a surrogate is used)
  - In supervised classification: cross-entropy loss as a surrogate of $I(X; Y)$

An Information-theoretic Perspective on ...

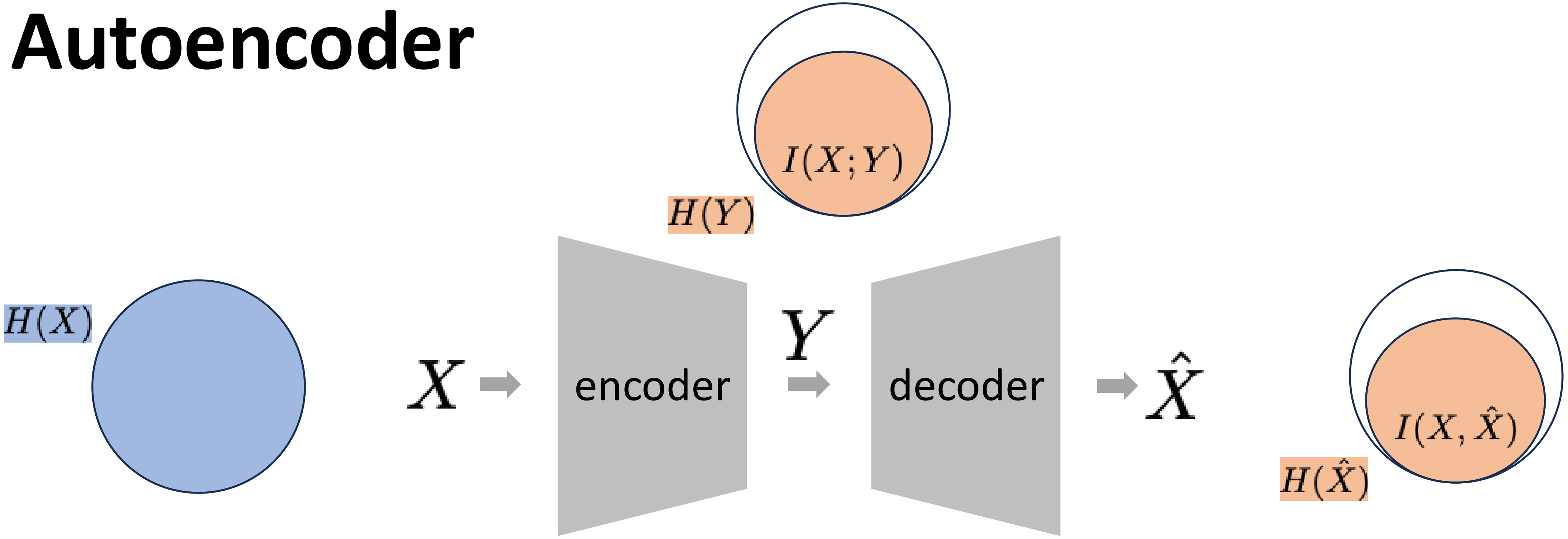
- Supervised Learning
- Self-supervised Learning: Autoencoding
- Self-supervised Learning: Contrastive Learning

Autoencoder



- The bottleneck limits the maximal information in the output
- Even a perfect decoder cannot recover more information (than what is passed through the bottleneck)

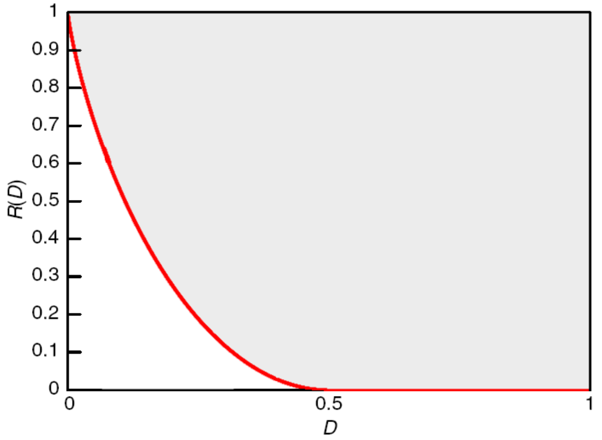
Autoencoder



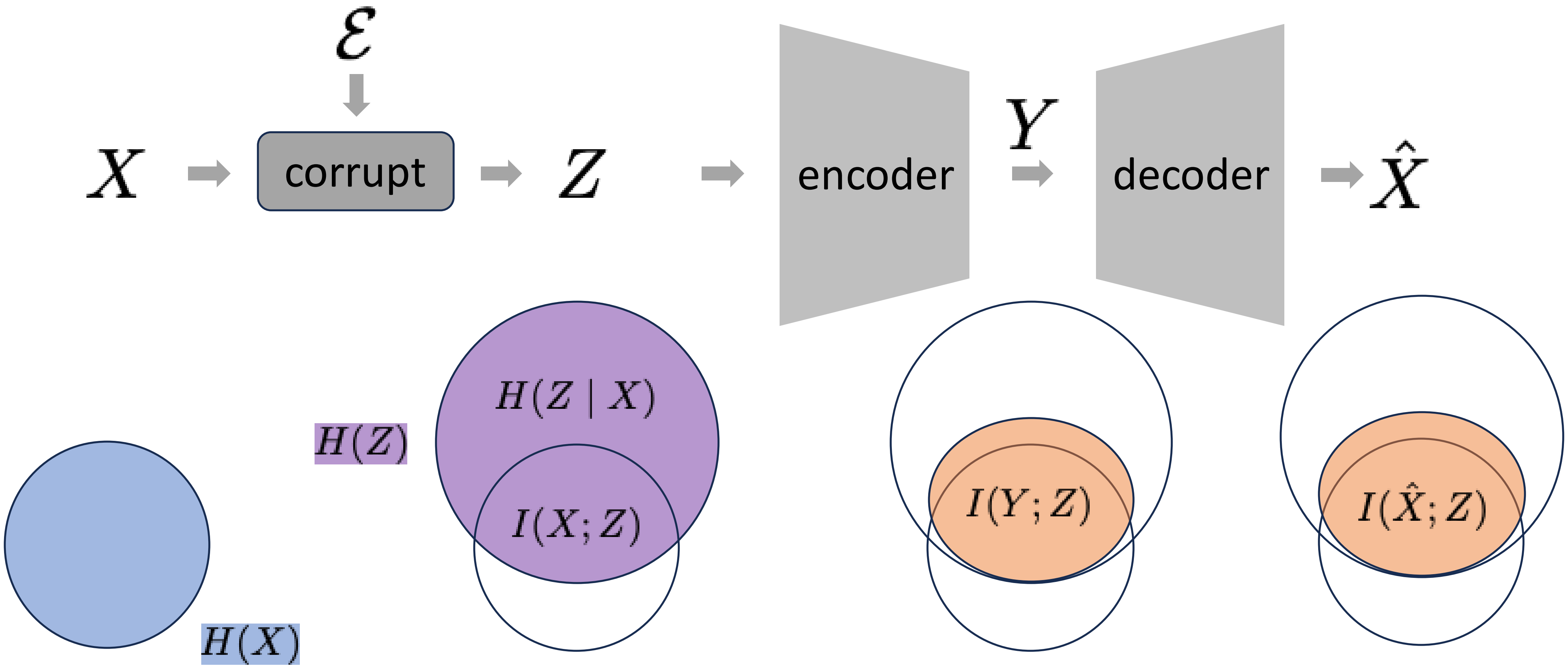
$$\text{“bits to encode } Y\text{”} \geq H(Y) \geq I(X;Y) \geq I(X;\hat{X}) \geq R(D)$$

As per Rate-distortion theory:

- Smaller distortion D ⇒ larger rate R(D))
- Minimizing distortion encourages more information retained
- Reconstruction error acts as a surrogate



Denoising / Masked Autoencoder



- Corruption discards some information
- Encoder can at best retain the remaining mutual information $I(X;Z)$

Vincent, et al., "Stacked Denoising Autoencoders: Learning Useful Representations in a Deep Network with a Local Denoising Criterion", JMLR, 2010

An Information-theoretic Perspective on ...

- Supervised Learning
- Self-supervised Learning: Autoencoding
- Self-supervised Learning: Contrastive Learning

Mutual Information Neural Estimation (MINE)

Problem setting:

- We have joint distribution $P(X, Y)$
  - unknown, but we can sample from it
- Goal: estimate mutual information $I(X; Y)$
- Approach: learn a neural net $T_\theta(x, y) : X \times Y \to \mathbb{R}$

Mutual Information:

$$I(X;Y) \triangleq D_{\mathrm{KL}}(\underbrace{p(x,y)}_{\text{joint}} \,||\, \underbrace{p(x)p(y)}_{\text{marginals}})$$

Belghazi et al., “Mutual Information Neural Estimation”, ICML 2018

Mutual Information Neural Estimation (MINE)

- Donsker-Varadhan's dual representation of KL divergence:

$$D_{\mathrm{KL}}(p \,||\, q) = \sup_{T:\mathcal{X}\to\mathbb{R}} \mathbb{E}_p[T] - \log \mathbb{E}_q[e^T]$$

- Mutual Information Neural Estimation (**MINE**):

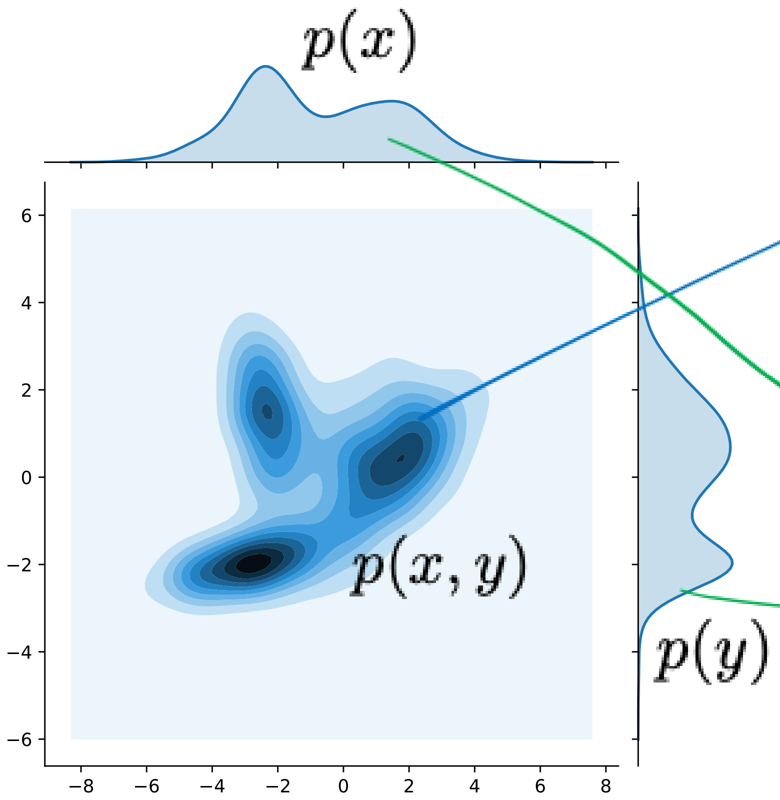
$$I(X;Y) = D_{\mathrm{KL}}(\underbrace{p(x,y)}_{\text{joint}} \,||\, \underbrace{p(x)p(y)}_{\text{marginals}}) \geq \sup_{T_\theta} \mathbb{E}_{\underbrace{p(x,y)}_{\text{joint}}}[T_\theta] - \log \mathbb{E}_{\underbrace{p(x)p(y)}_{\text{marginals}}}[e^{T_\theta}]$$

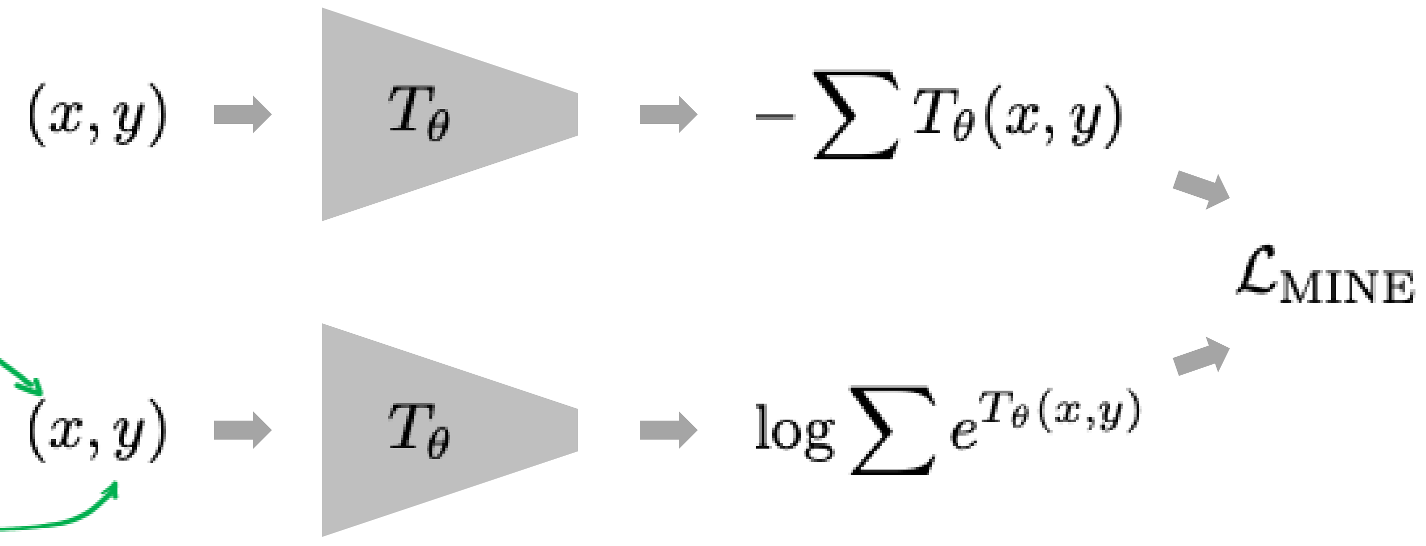
minimize loss: $\mathcal{L}_{\mathrm{MINE}}(\theta) = -\mathbb{E}_{p(x,y)}[T_\theta] + \log \mathbb{E}_{p(x)p(y)}[e^{T_\theta}]$

Belghazi et al., "Mutual Information Neural Estimation", ICML 2018

Mutual Information Neural Estimation (MINE)

$$\mathcal{L}_{\mathrm{MINE}}(\theta) = -\underbrace{\mathbb{E}_{p(x,y)}}_{\text{joint}}[T_\theta] + \log \underbrace{\mathbb{E}_{p(x)p(y)}}_{\text{marginals}}[e^{T_\theta}]$$





- Network has **two** inputs
- Network has a **scalar** output

Belghazi et al., "Mutual Information Neural Estimation", ICML 2018

InfoNCE (Noisy Contrastive Estimation)

$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E}\left[\log \frac{\exp(f(x) \cdot g(y^+)/\tau)}{\sum_{j=0}^{N} \exp(f(x) \cdot g(y_j)/\tau)}\right]$$

- x : **query** (anchor)
- y^+: a "**positive**" sample
- y_j : candidates (1 "positive", N "**negatives**")
- f, g : networks with **vector** output
- τ : temperature

What are positives and negatives?

InfoNCE (Noisy Contrastive Estimation)

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Relation to MINE:

$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E}\Big[f(x)\cdot g(y^+)/\tau\Big] + \mathbb{E}\Big[\log \sum_j \exp(f(x)\cdot g(y_j)/\tau)\Big]$$

$$\mathcal{L}_{\text{MINE}}(\theta) = -\mathbb{E}_{p(x,y)}[T_\theta] + \log \mathbb{E}_{p(x)p(y)}[e^{T_\theta}]$$

van den Oord et al., "Representation Learning with Contrastive Predictive Coding", arXiv 2018

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- f, g : accept a single input variable
- $T = f\cdot g/\tau$: restricted family of functions

van den Oord et al., “Representation Learning with Contrastive Predictive Coding”, arXiv 2018

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- **InfoNCE**: expectation over finite-sample log-sum-exp
- **MINE**: log of expectation (but approx. by log-sum-exp in practice)

van den Oord et al., "Representation Learning with Contrastive Predictive Coding", arXiv 2018

InfoNCE (Noisy Contrastive Estimation)

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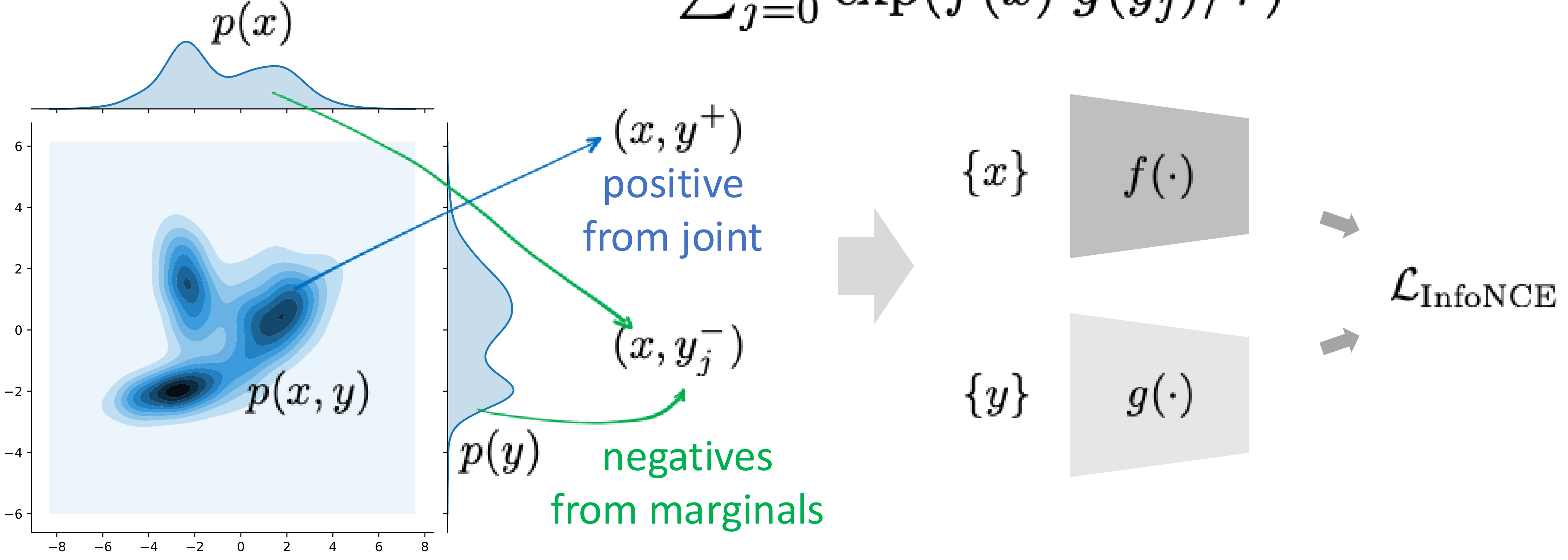
$$\mathcal{L}_{\text{MINE}}(\theta) = -\mathbb{E}_{p(x,y)}[T_\theta] + \log \mathbb{E}_{p(x)p(y)}[e^{T_\theta}]$$

- **"positive"** pair**:** from **joint** distribution
- "**negative**" pairs: from product of **marginals**

van den Oord et al., "Representation Learning with Contrastive Predictive Coding", arXiv 2018

InfoNCE (Noisy Contrastive Estimation)

$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E}\Big[\log \frac{\exp(f(x)\cdot g(y^+)/\tau)}{\sum_{j=0}^{N} \exp(f(x)\cdot g(y_j)/\tau)}\Big]$$



van den Oord et al., “Representation Learning with Contrastive Predictive Coding”, arXiv 2018

InfoNCE (Noisy Contrastive Estimation)

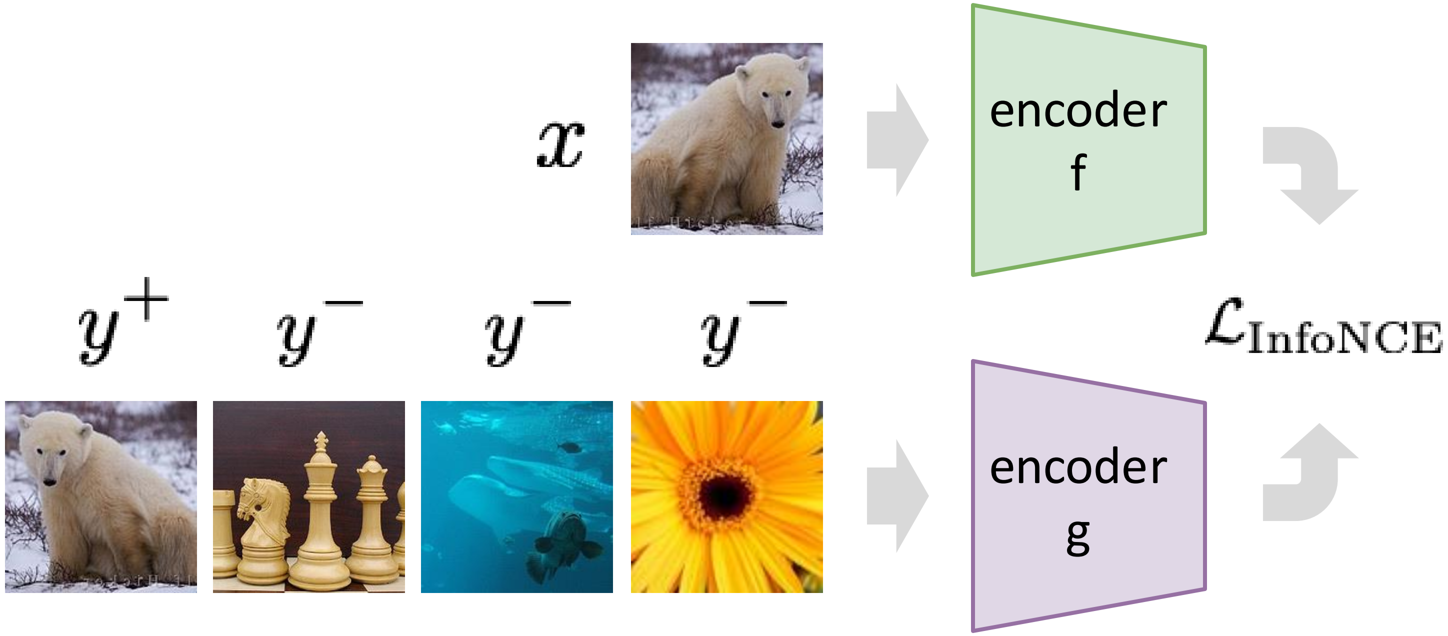
Estimating Mutual Information via InfoNCE:

$$I(X;Y) \geq \log(N) - \mathcal{L}_{\text{InfoNCE}}$$

- $\mathcal{L}_{\text{InfoNCE}}$ provides a lower bound on Mutual Information (up to a known constant $\log(N)$)
- Minimizing $\mathcal{L}_{\text{InfoNCE}}$ is to maximize the lower bound
- Bound tightens as N increases

van den Oord et al., “Representation Learning with Contrastive Predictive Coding”, arXiv 2018

Case Study: Contrastive Learning



Contrastive Learning: maximizing **Mutual Information** between an image and its augmented views.

- $x \sim p_{\text{im}}$, $y \sim p_{\text{im}}$: image distribution
- joint: $p(x, y^+) = p(y^+|x)p(x)$ where $p(y^+|x)$ is augmentation
- margins: $p(x)p(y)$, two random images

He et al. "Momentum Contrast for Unsupervised Visual Representation Learning", CVPR 2020
Chen et al. "A Simple Framework for Contrastive Learning of Visual Representations", ICML 2020

[Re-Recap] What are "good" representations?

We may want to answer:

- retain *what* information about the input?
- discard *what* information irrelevant to the tasks?
- compress *what* information under limited capacity?

still open questions ...

A note before we go ...

- Deep learning research is largely **empirical**

- Problem solving by **trial-and-error**
  - “human gradient descent”
  - with “reward” from experiments

- Theories provide good **motivations**
  - but evidence is needed

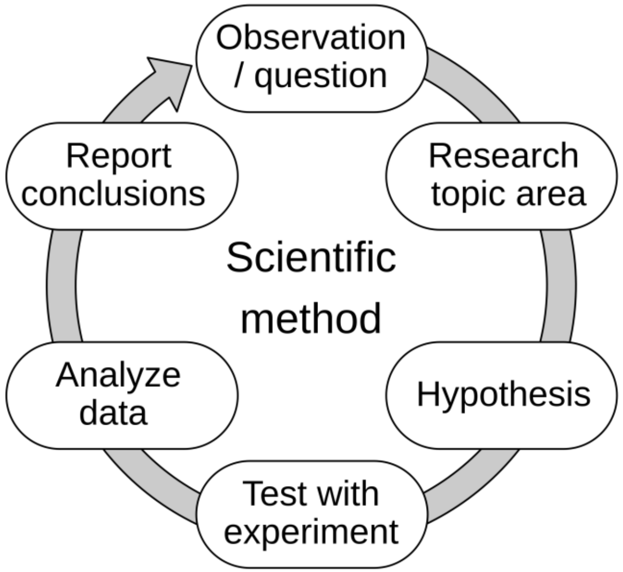


figure: https://en.wikipedia.org/wiki/Scientific_method

References

- Lecture Notes, 6.7480, 6.441, MIT. Y Polyanskiy
- "*Information Theory and Network Coding*", R Yeung. Springer, 2008
- "*The information bottleneck method*", Tishby et al., 2000
- "*Extracting and Composing Robust Features with Denoising Autoencoders*", Vincent et al., ICML, 2008
- "*Stacked Denoising Autoencoders: Learning Useful Representations in a Deep Network with a Local Denoising Criterion*", Vincent et al., JMLR, 2010
- "*Mutual Information Neural Estimation*", Belghazi et al., ICML 2018
- "*Representation Learning with Contrastive Predictive Coding*", van den Oord et al., arXiv 2018
- "*Learning deep representations by mutual information estimation and maximization*", R Hjelm et al., ICLR 2019

Overview: This Lecture

- Basics of Information Theory

- Modeling Information with Neural Networks